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1 D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-
  PIKAN_Official\experiments\Volterra1D_MCPI.py
2 The network model is cPIKAN
3 Layers: ModuleList(
4   (0): ChebyKANLayer(in_features=1, out_features=10, degree=3)
5   (1): ChebyKANLayer(in_features=10, out_features=10, degree=3)
6   (2): ChebyKANLayer(in_features=10, out_features=1, degree=3)
7 )
8 Total trainable parameters: 480
9 Model on device: cuda:0
10 Start training...
11 Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
12 Training: 100%|██████████| 2000/2000 [00:09<00:00, 216.62epoch/s, Loss=5.
  09e-06, LR=1.0e-03, Best=5.09e-06]
13 Training finished. Best loss: 3.85e-06. Model saved to results/best_model.pth
14 Loss history saved to results/loss_history.npy
15 Training time: 10.010724306106567s
16 Relative loss: 0.035479288548231125
17 L2 loss: 0.02257551997900009
18 Parameter containing:
19 tensor([[[ 0.0771,  0.1370,  0.4739, -0.0747],
20          [-0.0326, -0.5383, -0.2157,  0.1876],
21          [-0.0758,  0.3000,  0.3420,  0.0309],
22          [-0.1162, -0.2016, -0.0834,  0.7621],
23          [ 0.1917, -0.1449, -0.5391, -0.0980],
24          [-0.6113,  0.1391,  0.3448, -0.0526],
25          [ 0.2840, -0.0104, -0.2636, -0.2327],
26          [ 0.0128,  0.3370, -0.3191, -0.3091],
27          [ 0.6658, -0.1665, -0.6247, -0.3617],
28          [ 0.3088, -0.3226, -0.5779, -0.1011]]], device='cuda:0',
29         requires_grad=True)
30 Parameter containing:
31 tensor([[[ -6.8716e-03,  2.2280e-01,  1.3933e-02, -1.9551e-01],
32           [ 4.9127e-02, -2.2382e-01, -6.7982e-02,  2.4300e-01],
33           [ 2.5583e-02,  1.3698e-01, -2.0444e-02, -1.0257e-01],
34           [ 1.1328e-02, -2.1983e-01, -2.7269e-02,  2.0361e-01],
35           [-2.8446e-02,  2.1520e-01,  4.9160e-02, -1.5141e-01],
36           [ 3.8862e-02, -1.9771e-01, -3.3839e-02,  1.9668e-01],
37           [ 2.1382e-02, -1.7782e-01, -2.0515e-02,  1.6246e-01],
38           [ 4.7621e-02, -2.1391e-01, -1.0577e-02,  1.5188e-01],
39           [-2.3400e-02,  1.9291e-01, -1.5884e-02, -2.4161e-01],
40           [-2.8920e-02,  2.3867e-01, -7.3657e-03, -1.7211e-01]],
41          [[ -4.3681e-02, -2.1316e-01,  5.1025e-02,  1.8679e-01],
42           [ 3.3675e-02,  2.0679e-01, -9.2367e-03, -1.5811e-01],
43           [ 4.0555e-02, -2.4645e-01, -1.1656e-02,  2.6089e-01],
44           [ 4.6866e-02,  2.0290e-01, -4.7954e-02, -2.2979e-01],
45           [-2.2666e-02, -2.0535e-01,  4.1559e-02,  1.7421e-01],
46           [-1.6395e-02,  1.6154e-01, -6.1081e-02, -1.6911e-01],
47           [ 4.7492e-02,  2.3011e-01,  6.7074e-04, -1.8080e-01],
48           [-2.1982e-02,  2.1081e-01, -2.0415e-02, -2.0262e-01],
49           [-5.5379e-03, -2.7904e-01, -4.8022e-03,  2.2951e-01],
50           [-6.0500e-02, -1.9999e-01,  6.0052e-02,  1.7695e-01]]],
51         requires_grad=True)

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52
53      [[-2.1194e-02,  1.6796e-01, -2.9284e-02, -1.8438e-01],
54      [ 8.4711e-02, -2.9361e-01, -6.1691e-02,  2.8430e-01],
55      [ 6.0389e-03,  1.0209e-01,  3.0779e-03, -4.1271e-02],
56      [ 2.1680e-02, -2.3066e-01,  6.5270e-03,  2.0743e-01],
57      [-8.4517e-02,  2.5598e-01,  6.1390e-02, -2.0817e-01],
58      [ 5.1681e-02, -1.5440e-01, -3.1599e-02,  2.1165e-01],
59      [ 2.6777e-02, -2.0857e-01, -1.8278e-02,  1.5726e-01],
60      [ 3.1710e-02, -2.0267e-01, -3.4288e-02,  1.4374e-01],
61      [-4.6553e-02,  2.2369e-01, -3.8025e-02, -2.1688e-01],
62      [-6.7294e-03,  1.9181e-01,  1.2214e-02, -2.0643e-01]],
63
64      [[-4.7260e-02, -7.0429e-02, -2.9329e-02, -6.7069e-03],
65      [ 9.4788e-02, -6.1801e-02,  2.5393e-02, -7.8717e-02],
66      [ 8.9023e-02, -1.4540e-01,  2.2922e-02,  1.1410e-01],
67      [-3.6588e-03,  1.0228e-01,  1.3066e-02, -8.1054e-02],
68      [-4.4961e-02, -1.6906e-03,  2.6820e-03,  2.7301e-02],
69      [ 3.6092e-02, -1.2998e-02, -2.4281e-02, -2.4076e-02],
70      [-1.2303e-03, -2.8423e-03, -3.7874e-02, -1.8480e-02],
71      [ 2.2342e-02,  1.3390e-01, -5.5866e-02, -1.7343e-01],
72      [-1.9603e-02, -7.5079e-03,  4.3181e-02,  1.4340e-02],
73      [-1.2436e-02, -3.6615e-02,  5.5879e-02,  4.3942e-02]],
74
75      [[-1.9290e-02, -6.7577e-02, -4.2773e-03,  3.9076e-02],
76      [-5.7927e-03,  1.4643e-01, -1.9544e-02, -1.3930e-01],
77      [ 7.9136e-03, -4.8432e-03, -3.4044e-02, -4.1463e-02],
78      [ 6.9459e-03,  1.6202e-02, -8.8956e-03, -4.0596e-02],
79      [-3.8791e-02, -6.3533e-02,  3.1151e-03,  8.3722e-02],
80      [ 7.6616e-03,  1.1992e-01, -2.2408e-02, -3.1079e-02],
81      [ 1.2765e-02,  3.4733e-02, -2.1076e-02, -8.9011e-02],
82      [ 4.0862e-03,  8.0346e-02, -4.3076e-02, -6.1555e-02],
83      [ 2.3829e-03, -7.6036e-02, -1.0100e-02, -8.0203e-03],
84      [-2.7468e-02, -6.9797e-02, -1.5719e-02,  8.5898e-02]],
85
86      [[-3.1731e-02,  4.3701e-02, -2.0318e-02, -2.0982e-02],
87      [ 4.6068e-02, -2.9386e-02,  6.6924e-02,  6.6577e-02],
88      [ 8.4762e-03, -1.7724e-02, -6.3955e-02,  1.0235e-02],
89      [ 1.2848e-02, -8.4032e-02,  7.0641e-02, -2.2606e-02],
90      [-2.4211e-02,  1.6399e-02, -2.6193e-02, -6.5473e-02],
91      [ 5.0109e-02, -2.2749e-03,  5.4931e-02,  3.7078e-02],
92      [ 2.9252e-02, -3.4946e-02,  8.4870e-03, -1.0011e-02],
93      [ 1.6490e-02, -5.2805e-02,  1.0310e-01, -2.4898e-02],
94      [-5.8120e-03,  1.7234e-02, -6.2277e-03, -1.8321e-02],
95      [-4.7644e-02,  8.8752e-03, -4.0904e-02, -1.8662e-02]],
96
97      [[ 3.0095e-02, -1.0950e-02,  7.1637e-02,  3.8115e-02],
98      [ 3.3539e-02,  5.6685e-02,  3.3181e-02, -1.0102e-01],
99      [-5.6577e-02,  3.9195e-02,  1.8175e-02,  1.5096e-03],
100     [-1.9287e-02,  1.2607e-02, -2.8482e-02, -1.5831e-02],
101     [-1.0363e-02, -3.0198e-02,  4.2444e-03, -9.8455e-03],
102     [ 7.3110e-03,  2.8254e-02,  2.9751e-04, -2.5883e-02],
103     [ 5.8372e-02,  3.5092e-02, -2.5022e-04, -5.0742e-02],
104     [ 4.1454e-02, -5.8574e-03, -7.6380e-02, -4.1156e-02],

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105      [ 5.9175e-03, -6.9111e-03,  5.0757e-02, -1.5949e-03],
106      [-6.5007e-02, -3.5675e-02,  1.7604e-02,  1.1687e-02]],
107
108      [[-9.8024e-02, -2.5036e-02,  2.4095e-02,  4.8624e-02],
109      [ 5.1061e-02,  1.0484e-01, -2.7998e-03, -2.8358e-02],
110      [-7.0558e-03,  2.4086e-02,  3.6963e-02, -7.5693e-02],
111      [ 7.2867e-02, -3.8628e-02, -6.5939e-02,  1.5050e-02],
112      [-4.4343e-02,  4.9629e-02,  5.2817e-02,  6.0268e-02],
113      [ 2.3260e-02,  2.9332e-02, -1.3397e-02, -1.7555e-02],
114      [-9.5745e-03,  5.0592e-02, -2.6590e-02, -1.3708e-02],
115      [ 3.9169e-02, -3.1617e-02, -5.4775e-02,  2.6096e-02],
116      [-5.0909e-02,  2.5573e-03,  4.2259e-02,  1.6316e-02],
117      [-4.9541e-02, -4.0824e-02,  5.3392e-03, -5.5113e-04]],
118
119      [[ 1.8766e-02, -3.0169e-02, -9.8082e-02, -1.0114e-02],
120      [ 4.5728e-02,  9.3311e-02,  2.7965e-01, -8.7287e-03],
121      [ 8.7491e-04,  5.9381e-02,  3.3724e-02,  8.7783e-03],
122      [-4.6805e-03,  8.9854e-03,  7.5879e-02, -5.2221e-02],
123      [-2.9940e-02, -4.8684e-02, -1.7843e-01,  6.8564e-03],
124      [-1.3742e-02,  3.2419e-02,  1.1583e-01, -3.1460e-02],
125      [ 4.8978e-02,  1.0281e-03,  1.4987e-01, -1.5316e-02],
126      [ 3.8627e-02,  3.1068e-02,  6.5046e-02, -8.9687e-02],
127      [ 2.9927e-02, -1.6501e-02, -6.3774e-02,  3.6318e-02],
128      [-2.1570e-02, -5.4109e-02, -1.2107e-01,  5.6730e-02]],
129
130      [[-9.8643e-03, -2.8026e-02,  4.5891e-02,  6.8624e-02],
131      [ 3.4713e-02,  1.2706e-01,  1.3822e-02, -1.5626e-01],
132      [ 3.4490e-02,  2.1738e-02, -3.0643e-02, -1.7528e-02],
133      [ 1.5741e-02,  7.2026e-02,  3.9466e-02, -4.0124e-02],
134      [-2.6093e-02, -5.6507e-02, -1.5172e-02,  5.6728e-02],
135      [ 4.2302e-02,  5.7390e-02, -1.4810e-03, -6.4481e-02],
136      [ 5.4814e-02,  4.3159e-02, -1.6624e-02, -2.8310e-02],
137      [ 3.1503e-02,  8.5972e-02,  1.2840e-01, -2.6461e-02],
138      [-8.6844e-02, -5.2687e-02, -4.2434e-02,  3.2130e-02],
139      [-1.9451e-02, -9.1112e-02, -1.1168e-02,  5.0636e-02]]],
140      device='cuda:0', requires_grad=True)
141 Parameter containing:
142 tensor([[[[-0.0055, -0.0155, -0.0293,  0.1358]],
143
144          [[ 0.0034,  0.0266, -0.0253, -0.0679]],
145
146          [[ 0.0329,  0.0526,  0.0190, -0.1004]],
147
148          [[ 0.0649,  0.0111, -0.0806, -0.0768]],
149
150          [[-0.0157, -0.0023, -0.0003,  0.0684]],
151
152          [[ 0.0340,  0.0161, -0.0221, -0.0990]],
153
154          [[ 0.0065,  0.0019,  0.0165, -0.0700]],
155
156          [[ 0.0077, -0.0177, -0.0457, -0.0882]],
157

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158      [[ 0.0539,  0.0008, -0.0509,  0.0557]],
159
160      [[-0.0275, -0.0237, -0.0168,  0.0515]]], device='cuda:0',
161      requires_grad=True)
162 =====
163      Layer (type:depth-idx)          Output Shape          Param #
164      =====
165      CPIKANModel                    [1, 1]                  --
166      |---ModuleList: 1-1              --                      --
167      |   |---0.weight                  |---40
168      |   |---1.weight                  |---400
169      |   |---2.weight                  |---40
170      |   |---ChebyKANLayer: 2-1        [1, 10]                 40
171      |   |   |---weight                |---40
172      |   |---ChebyKANLayer: 2-2        [1, 10]                 400
173      |   |   |---weight                |---400
174      |   |---ChebyKANLayer: 2-3        [1, 1]                  40
175      |   |   |---weight                |---40
176      =====
177      Total params: 480
178      Trainable params: 480
179      Non-trainable params: 0
180      Total mult-adds (Units.MEGABYTES): 0.00
181      =====
182      Input size (MB): 0.00
183      Forward/backward pass size (MB): 0.00
184      Params size (MB): 0.00
185      Estimated Total Size (MB): 0.00
186      =====
187
188      进程已结束, 退出代码为 0
189

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